

EXTENDS *Naturals, Sequences, FiniteSets*CONSTANTS *Keys, Values, InitVal, Workload* $TxnIds \triangleq \text{DOMAIN } Workload$ $InitTx \triangleq 0$ $AllTxns \triangleq TxnIds \cup \{InitTx\}$ $IsoLevels \triangleq \{“RA”, “CC”, “PC”, “PSI”, “SI”, “SER”\}$

VARIABLES *client_dep,*
client_last_commit,
router_ct,
shard_ct,
tx_status,
tx_snapshot,
tx_checkset,
tx_buffer,
tx_commit_time,
pc,
store,
op_seq

$all_vars \triangleq$
 $\langle$ *client_dep,*
client_last_commit,
router_ct,
shard_ct,
tx_status,
tx_snapshot,
tx_checkset,
tx_buffer,
tx_commit_time,
pc,
store,
op_seq
 $\rangle$

1. OPERATIONAL PROTOCOL

$Init \triangleq$
 LET $Sessions \triangleq \{Workload[t].session : t \in TxnIds\}$
 IN $\wedge client_dep = [s \in Sessions \mapsto 0]$
 $\wedge client_last_commit = [s \in Sessions \mapsto 0]$